

VARIABLES

ANALYSIS

Extracting insight from features-pair

Objectives:

- Techniques to Analyze Two Numerical Variables
- Techniques to Analyze Two Categorical Variables
- Techniques to Analyze a Numerical Variable and Categorical Variables

Best Techniques to Analyze Two Numerical Variables

Analyzing **two numerical variables together** helps you understand *relationships, trends, patterns, correlations, and dependencies*. Below is a complete list of the most widely used methods in Data Analysis and Data Science.

1 Scatter Plot

✓Use When:

You want to visually inspect the relationship between two continuous variables.

✓Insight It Provides:

- Linear relationships
- Non-linear relationships
- Clusters
- Outliers
- Strength and direction of trends

✓Why It Works:

Scatter plots reveal patterns that statistics alone cannot show.

2 Correlation Coefficient (Pearson)

✓Use When:

You want to measure **how strongly** two variables move together.

✓Insight It Provides:

- Strength (weak → strong)
- Direction (positive or negative)
- Linear relationship only

✓Why It Works:

Pearson correlation quantifies the **degree** of linear dependency.

Example:

- $r = +0.90 \rightarrow$ strong positive relationship
- $r = -0.35 \rightarrow$ weak negative relationship

3 Spearman Rank Correlation

✓Use When:

- Data is not normally distributed
- Relationship is monotonic but non-linear
- Variables contain outliers

✓Insight It Provides:

Measures monotonic relationships (increasing or decreasing).

✓Why It Works:

Uses ranks instead of raw values → robust to non-parametric data.

4 Regression Analysis (Simple Linear Regression)

✓Use When:

You want to *predict* one variable from another or understand *cause-effect tendencies*.

✓Insight It Provides:

- Slope (change in Y given change in X)
- Intercept
- Goodness of fit (R^2)
- Direction + strength
- Practical predictive formulas

✓Why It Works:

It quantifies how much one variable changes when the other changes.

5 Joint Plot

✓Use When:

You want to analyze **scatter + distribution** in one chart.

✓Insight It Provides:

- Relationship (via scatter)
- Marginal distributions (via histograms/kde)

✓Why It Works:

It helps you understand whether the relationship is driven by specific ranges of values.

6 Hexbin Plot

✓Use When:

You have **many data points** (thousands or millions), and scatter plots overlap too much.

✓Insight It Provides:

- Density of points

- Hotspots of relationships

✓Why It Works:

Bins the plane into hexagons → reduces overplotting.

7 2D KDE Plot (Kernel Density Estimation)

✓Use When:

You want a smooth, continuous estimate of joint density.

✓Insight It Provides:

- Areas where data is most concentrated
- Shape of distribution pairs

✓Why It Works:

It smooths data across space, helping you understand density distributions.

8 Covariance

✓Use When:

You need to know whether variables move together (but not how strongly).

✓Insight It Provides:

- Positive covariance → move in same direction
- Negative covariance → move in opposite direction
- Zero covariance → no linear relationship

✓Why It Works:

Forms the foundation for correlation but not scale-independent.

9 ☐ Pairwise Distance / Clustering Based on Numerical Variables

✓ Use When:

You believe two variables form patterns or clusters of values.

✓ Insight It Provides:

- Natural grouping patterns
- Outlier clusters
- Variable relationships at cluster level

✓ Why It Works:

Clustering reveals hidden structure beyond simple correlations.

☐ Binning + Grouped Statistics

✓ Use When:

You want insight into how variable relationships change across intervals.

✓ Insight It Provides:

- Averages, medians, variances per bin
- Non-linear patterns
- Step-like relationships

✓ Why It Works:

Transforms continuous variables into interpretable segments.

☐ Summary: When to Use What

Technique	Best For	Insight
Scatter Plot	First check	Visual pattern
Pearson Correlation	Linear relationships	Strength & direction
Spearman Correlation	Non-linear monotonic	Ranked correlation
Regression	Predictive relationships	Slope, intercept, R^2
Jointplot	Scatter + distribution	Combined view
Hexbin/KDE	Large datasets	Density visualization
Covariance	Basic co-movement	Relationship direction
Clustering	Hidden patterns	Natural groups
Binning	Non-linear insights	Local behavior

Best Techniques for Analyzing a Pair of Categorical Variables

When you have **two categorical variables**, the goal is usually to understand:

- Whether the categories are **related or independent**
- How often certain category combinations occur
- Which combinations are **dominant, rare, or unusual**
- Patterns that help with **prediction, segmentation, or quality checks**

Below are the core techniques.

1 Contingency Table (Cross-Tabulation)

Why use it?

This is the **foundation** of analyzing categorical pairs. It shows how categories intersect.

What it gives:

- Counts of every combination
- Percentages (row %, column %, overall %)
- Identification of dominant or unusual category pairs

Insights:

- Which category combinations are frequent
 - Distribution differences
 - Basic relationships
-

2 Stacked or Grouped Bar Charts

Why use it?

They visually compare distributions across categories, making differences easy to spot.

Variations:

- **Stacked bar chart**
- **100% stacked bar chart** → compares proportions
- **Grouped/clustered bar chart**

Insights:

- Clear visual comparison between categories
 - Highlights disproportionate representation
-

3 Heatmap (From Contingency Table)

Why use it?

It adds color intensity, making patterns “pop out.”

Insights:

- Identifies strong associations quickly
 - Detects hot spots and anomalies
 - Easy interpretation for non-technical audiences
-

4 Chi-Square Test of Independence

Why use it?

Statistical test used to check whether two categorical variables are **related** or **independent**.

What it answers:

- Are the variables dependent?
- Is the association statistically significant?

Insights:

- Confirms whether patterns in the table are meaningful
 - Useful for surveys, risk assessment, market analysis
-

5 Cramér's V (Strength of Association)

Why use it?

The Chi-Square test tells you **if** an association exists.
Cramér's V tells you **how strong** it is.

Scale:

- 0 → no association
- 1 → perfect association

Insights:

- Quantifies association strength
 - Helps prioritize variables for modeling
-

6 Mosaic Plots

Why use it?

Visual representation of the contingency table where tile sizes represent frequency.

Insights:

- Shows how proportions differ across categories
 - Good for spotting imbalance or skew
 - Serves as a visual counterpart to Chi-Square
-

7 Association Rules (Advanced)

Useful if you have **many categorical variables**, but can work with pairs too.

Why use it?

It uncovers rules like:

“Customers who choose X usually choose Y.”

Metrics:

- **Support** — how common the combination is
- **Confidence** — how likely Y follows X
- **Lift** — strength of the rule

Insights:

- Hidden patterns that aren't obvious
 - Useful for marketing, recommendation systems
-

8 Clustering of Categorical Variables (Advanced)

Why use it?

If categories are complex (e.g., dozens of levels), clustering can group similar patterns.

Techniques:

- k-modes
- Hierarchical clustering on contingency frequencies
- MCA-based clustering

Insights:

- Identifies relationships between many category combinations
 - Helps simplify complex categorical structures
-

9 Multiple Correspondence Analysis (MCA)

Why use it for pairs?

With only two categorical variables, MCA simplifies and visualizes relationships.

Insights:

- Shows how categories relate in 2-D space
- Highlights clusters or distances
- Useful for dimensional reduction

☐ Lift and Odds Ratios (Prediction-Focused)

Why use it?

Helps measure how much one category boosts the likelihood of another.

Insights:

- Good for risk modeling
- Shows predictive power
- Helps in marketing and classification tasks

☐ Summary Table

Technique	Purpose	Why Useful
Contingency Table	Counts relationships	Foundation of all analysis
Bar Charts	Compare distributions	Great for presentations
Heatmap	Visualize intensity	Reveals strong patterns
Chi-Square Test	Check independence	Statistical validation
Cramér's V	Measure strength	Quantifies relationship
Mosaic Plot	Visual proportions	Great diagnostic tool
Association Rules	Find hidden patterns	Good for large datasets

MCA	Visualize associations	Dimensionality reduction
Lift/Odds Ratios	Predictive relationships	Useful in modeling

Best Techniques for Analyzing a Categorical + Numerical Variable Pair

The goal is usually to answer:

- Does the numerical value differ across categories?
- Which categories have the highest/lowest values?
- Is the category predictive of the numerical variable?
- Are there outliers or variations within categories?

Below are the top techniques.

1 ☐ Group-by Summary Statistics (Mean, Median, Std, etc.)

Why use it?

This is the **foundation**. You compute common statistics grouped by category.

What to compute:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum

- Count

Insights gained:

- Which categories have higher or lower values
 - Variation within each category
 - Central tendencies (mean/median differences)
-

2 Box Plots

Why use it?

Boxplots are **ideal** for comparing numerical distributions across different categories.

Insights gained:

- Which category has the highest/lowest median
- Spread of data (IQR)
- Presence of outliers
- Distribution shape differences

Why chosen:

Quick, intuitive, and non-technical audiences understand them easily.

3 Violin Plots

Why use it?

A violin plot shows the **full distribution** of numerical values per category, unlike a box plot.

Insights gained:

- Shape of the underlying distribution
 - Multiple peaks (bimodality)
 - Density differences between categories
-

4 Bar Charts of Means or Medians

Why use it?

Simple and very effective for presentations.

Insights gained:

- Average values per category
- Easy comparison
- Shows ranking of categories

Why chosen:

Executives prefer this over more complex charts.

5 Strip Plot / Swarm Plot

Why use it?

Shows **each individual data point**, grouped by category.

Insights gained:

- Data clustering patterns
 - Outliers
 - Gaps or grouping in the data
-

6 ANOVA (Analysis of Variance)

Why use it?

A statistical test that checks whether **the numerical variable differs significantly across categories**.

Interpretation:

- **p-value < 0.05** → categories have statistically different means
- **p-value > 0.05** → no evidence of meaningful difference

Insights gained:

- Confirms if visual differences are real or random
 - Helps in predictive modeling and feature selection
-

7 ☐ Effect Size Measures (η^2 or Cohen's d)

Why use it?

Statistical significance does not tell us **strength**.

Effect size measures tell how much the category influences the numerical value.

Insights gained:

- Practical significance
 - Whether the effect is weak, moderate, or strong
-

8 ☐ T-tests (for 2 Categories Only)

Why use it?

If your categorical variable has only **two levels**, a t-test is powerful and simple.

Insights gained:

- Whether the two category groups differ in mean
 - Strength and direction of difference
-

9 ☐ Regression with One-Hot Encoding

Why use it?

Allows you to measure how each category influences the numeric variable **while controlling for variance**.

Insights gained:

- Predictive relationships
 - Category impact on the scale of the numeric variable
 - Model coefficients show how much each category shifts the value
-

☐ Distribution Comparison Across Categories

Use KDE (kernel density estimation) or histograms for each category.

Why use it?

Shows how the underlying numeric distribution differs.

Insights gained:

- Whether categories overlap
- Whether categories create separation between numerical values

☐ **Summary Table**

Technique	Purpose	Why It's Useful
Group-by Summary Stats	Compare central tendencies	Foundation of categorical–numeric analysis
Box Plot	Compare distributions	Shows outliers, spread, medians
Violin Plot	Shows full distribution shape	Great for advanced insights
Bar Chart of Means	Compare average values	Great for storytelling
Strip / Swarm Plot	Individual data points	Shows raw patterns & clustering
ANOVA	Statistical test	Determines significance
Effect Size	Strength of effect	Adds practical understanding
T-test	Two-category comparisons	Simple but powerful
One-hot Regression	Modeling relationship	Shows category contributions
KDE / Histograms	Distribution comparison	Reveals overlap or separation

Exploratory Data Analysis (Most Common & Most Useful)

A. Grouped Descriptive Statistics

Compute numerical summaries **per category**, such as:

- mean
median
- standard deviation
- min/max
- Quantiles

Example (Python):

```
df.groupby("category")["value"].describe()
```

Insight example:

“Category A has much higher average sales than B and C.”

B. Boxplots

A boxplot shows:

- median
- interquartile range
- outliers
- distribution shape for each category

Use when:

You want to compare numeric distributions across categories.

C. Violin Plots

Like boxplots but also show **full distribution density**.

Use when:

You want deeper insight into distribution shape (multi-modal, skewed, etc.).

D. Strip / Swarm / Dot Plots

Show individual data points inside each category.

Use when:

Data is small/moderate and you want detail, not just summaries.

E. Bar Plot of Means (With CI)

Plot category means plus 95% confidence intervals.

Use when:

You want a quick overview of differences between categories.

F. Histograms by Category

Overlay or facet histograms for each category.

Use when:

You want to see distribution shape differences.

✓2. Statistical Tests (To check if differences are *significant*)

A. ANOVA (One-Way)

Use when:

- numerical outcome
- categorical variable has **>2 categories**

Tests: Are all category means equal?

```
from scipy.stats import f_oneway
f_oneway(df[df.cat=="A"].value, df[df.cat=="B"].value, ...)
```

B. t-Test

Use when:

- categorical variable has **2 categories**
-

C. Kruskal-Wallis

Non-parametric version of ANOVA.

Use when:

- data is not normally distributed
- variance is unequal

D. Mann-Whitney U

Non-parametric version of t-test.

E. Effect Size Measures

Important for interpretation:

- Cohen's d (2 groups)
- Eta-squared (ANOVA)
- Kruskal effect size

✓3. Relationship Strength & Metrics

A. Correlation Ratio (η)

Measures how strongly a numerical variable depends on a categorical variable.

Better than correlation for mixed types.

B. Mutual Information

Captures non-linear relationships.

Example:

```
from sklearn.metrics import mutual_info_regression
```

✓4. Model-Based Approaches (For Deeper Insight)

A. Decision Trees

Feature importance reveals how important the categorical variable is in predicting the numerical variable.

B. Linear Regression with Categorical Encoding

Encode category using:

- One-Hot (dummy)
- Target encoding (advanced)
- Ordinal encoding (if categories have order)

Example:

`value ~ C(category)`

This is the regression form of ANOVA.

C. Random Forest Feature Importance

Shows whether the categorical variable helps predict numerical outcome.

✓5. Additional Advanced Insights

A. Probability Density Comparison (Kernel Density Plots)

Shows smooth distribution curves per category.

B. Cumulative Distribution Comparison (ECDF curves)

Useful to compare entire distributions (not just central tendency).

C. Bootstrapping

Estimate:

- difference in means
- difference in medians
- confidence intervals

Great for non-normal data.

★ Summary Table

Goal	Best Techniques
Compare distributions	Boxplot, violin, KDE
Compare means	Group statistics, barplot with CI
Test significance	t-test, ANOVA, Kruskal
Measure relationship	Correlation ratio (η), Mutual Information
Model insight	Regression, decision trees, random forest
Visual granular detail	Strip plot, swarm plot

Step-by-step workflow

Step 0: Import Libraries and Sample Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.feature_selection import mutual_info_regression

# Sample dataset
data = {
    "Category": np.random.choice(['A', 'B', 'C'], size=100),
    "Value": np.concatenate([
        np.random.normal(50, 10, 30),
        np.random.normal(60, 12, 35),
        np.random.normal(55, 8, 35)
    ])
}

df = pd.DataFrame(data)
```

Step 1: Basic Descriptive Statistics

```
# Overall stats
print(df["Value"].describe())

# Stats by category
print(df.groupby("Category")["Value"].describe())
```

Insight: Mean, median, min/max, and spread per category.

Step 2: Visual Exploration

A. Boxplot

```
plt.figure(figsize=(8,6))
sns.boxplot(x="Category", y="Value", data=df)
plt.title("Boxplot of Value by Category")
plt.show()
```

B. Violin Plot

```
plt.figure(figsize=(8,6))
sns.violinplot(x="Category", y="Value", data=df)
plt.title("Violin Plot of Value by Category")
plt.show()
```

C. Strip/Swarm Plot

```
plt.figure(figsize=(8,6))
sns.swarmplot(x="Category", y="Value", data=df, color='black')
```

```
plt.title("Swarm Plot of Value by Category")
plt.show()
```

D. Barplot with Confidence Interval

```
plt.figure(figsize=(8,6))
sns.barplot(x="Category", y="Value", data=df, ci=95)
plt.title("Mean Value by Category with 95% CI")
plt.show()
```

Step 3: Statistical Testing

A. ANOVA (if >2 categories)

```
groups = [df[df["Category"]==c]["Value"] for c in df["Category"].unique()]
f_stat, p_value = stats.f_oneway(*groups)
print("ANOVA F-statistic:", f_stat)
print("ANOVA p-value:", p_value)
```

Interpretation:

- $p\text{-value} < 0.05 \rightarrow$ means are significantly different.

B. Kruskal-Wallis (Non-parametric alternative)

```
h_stat, p_value = stats.kruskal(*groups)
print("Kruskal-Wallis H-statistic:", h_stat)
print("p-value:", p_value)
```

Step 4: Relationship Measures

A. Correlation Ratio (η)

```
def correlation_ratio(categories, values):
    categories = np.array(categories)
    values = np.array(values)
    cat_means = [values[categories == cat].mean() for cat in np.unique(categories)]
    n_total = len(values)
    grand_mean = values.mean()
    ss_between = sum([len(values[categories==cat]) * (m - grand_mean)**2 for cat, m in
zip(np.unique(categories), cat_means)])
    ss_total = sum((values - grand_mean)**2)
    return np.sqrt(ss_between / ss_total)
```

```
eta = correlation_ratio(df["Category"], df["Value"])
print("Correlation Ratio ( $\eta$ ):", eta)
```

B. Mutual Information

```
mi = mutual_info_regression(pd.get_dummies(df["Category"]), df["Value"])
print("Mutual Information:", mi)
```

Step 5: Model-Based Insight

A. Linear Regression

```
import statsmodels.api as sm
df_encoded = pd.get_dummies(df, drop_first=True) # Encode categories
```

```
X = df_encoded.drop("Value", axis=1)
y = df_encoded["Value"]

X = sm.add_constant(X) # Add intercept
model = sm.OLS(y, X).fit()
print(model.summary())
```

B. Decision Tree Feature Importance

```
from sklearn.tree import DecisionTreeRegressor

X = pd.get_dummies(df["Category"])
y = df["Value"]
tree = DecisionTreeRegressor()
tree.fit(X, y)
print("Feature importance:", tree.feature_importances_)
```

Step 6: Optional Advanced Visualizations

A. KDE / Distribution per category

```
plt.figure(figsize=(8,6))
sns.kdeplot(data=df, x="Value", hue="Category", fill=True)
plt.title("Density Plot of Value by Category")
plt.show()
```

B. ECDF Curves

```
import numpy as np

for cat in df["Category"].unique():
    x = np.sort(df[df["Category"]==cat]["Value"])
    y = np.arange(1, len(x)+1) / len(x)
    plt.step(x, y, label=cat)
plt.title("ECDF by Category")
plt.xlabel("Value")
plt.ylabel("ECDF")
plt.legend()
plt.show()
```

✓ Workflow Summary

1. **Check basic stats** (overall & by category)
2. **Visualize distributions** (boxplot, violin, swarm, barplot)
3. **Test statistical significance** (ANOVA, t-test, Kruskal-Wallis)
4. **Measure relationships** (correlation ratio, mutual information)
5. **Model insights** (regression, decision trees, feature importance)
6. **Advanced visualizations** (KDE, ECDF, bootstrapping if needed)